



Introduction & Motivation

Singapore's trade network involves a diverse range of partner countries with different trade levels, structures, and long-term movement patterns.

As these relationships evolve over time, simple country-by-country comparison is often insufficient to reveal broader trends and meaningful groupings.

Motivation:

This project is motivated by the **need to better understand Singapore's trade partner landscape** through visual analytics, combining exploratory & confirmatory analysis, forecasting, and clustering to uncover **structural and temporal patterns** across countries.

R shiny packages

The project was prepared using R programming as the main courseware and maintain using Quarto for the main website (hosted on Netlify). R Shiny Dashboard was used to build the web application.

Main R packages:

fable, feasts: auto-ETS

Leaflet, rnatuarearth: Interactive Map

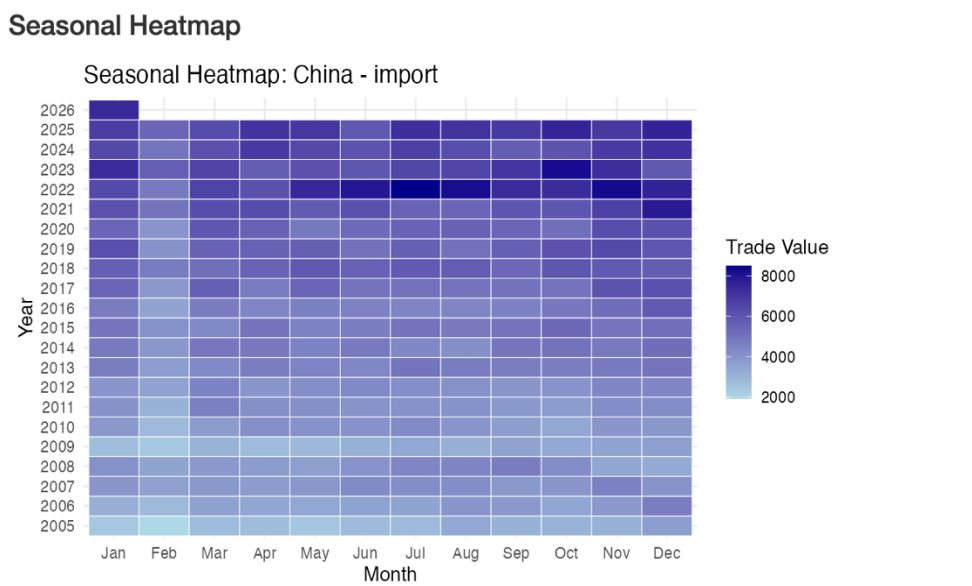
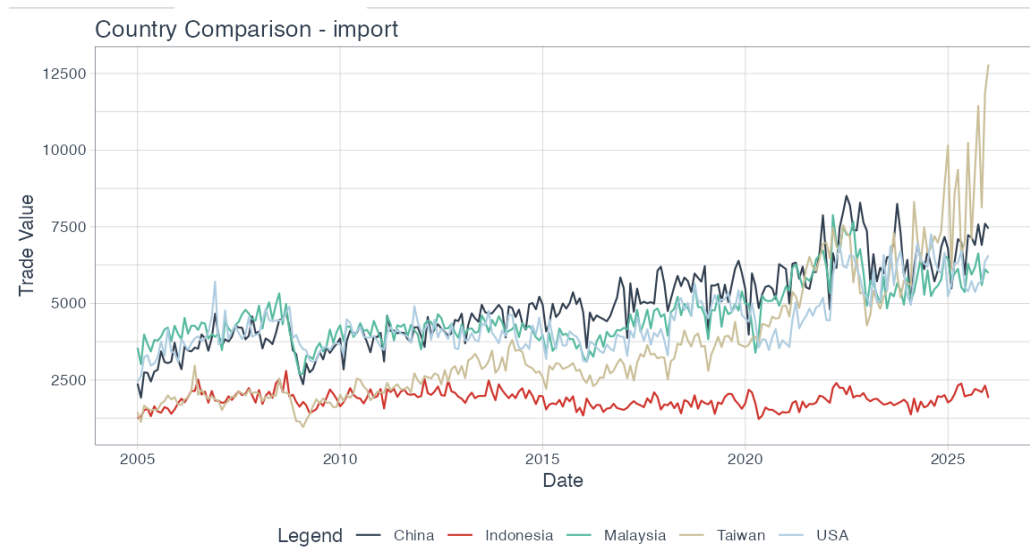
ggplot2: data visualisation

lubridate: time-based data handling and feature extraction

timetk: time series exploration and visual analytics



Exploratory & Confirmatory Analysis



Significant disparities exist across countries, with some exhibiting consistently higher trade volumes, suggesting strong trade concentration among key partners.

A recurring seasonal pattern is observed, with certain months consistently showing higher trade values, indicating potential cyclical demand or policy-driven effects.

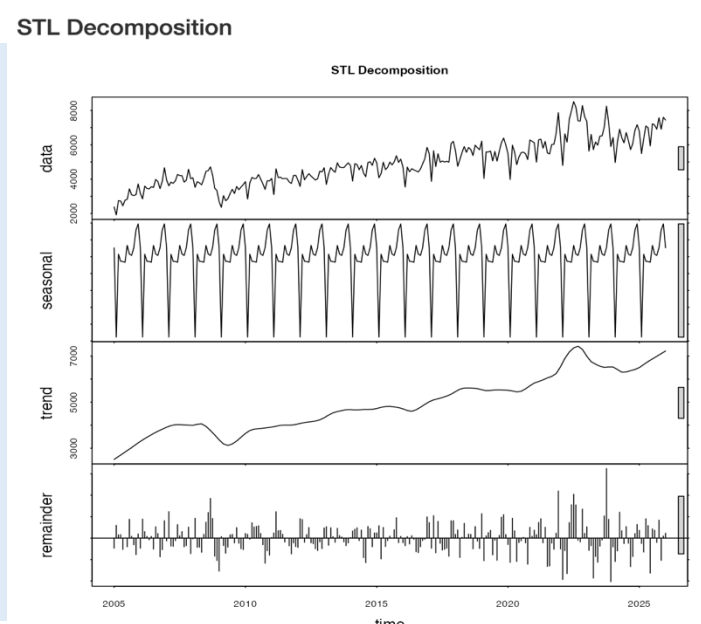
- Users can dynamically filter countries to explore country-specific trade patterns and compare temporal behaviours across regions.

Description: Decomposes trade data into trend, seasonal, and irregular components.

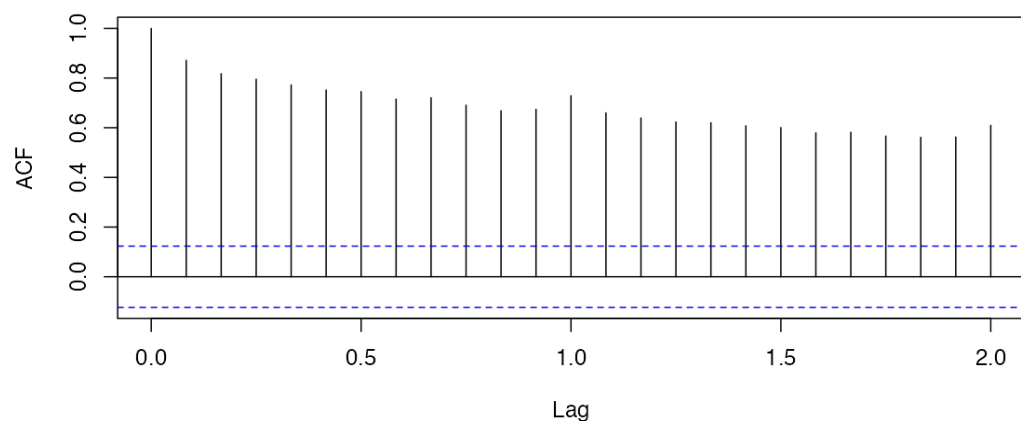
Trend: Shows long-term movement and structural changes over time.

Seasonality: Captures recurring monthly patterns in trade activity.

Remainder: Represents irregular fluctuations and potential anomalies.



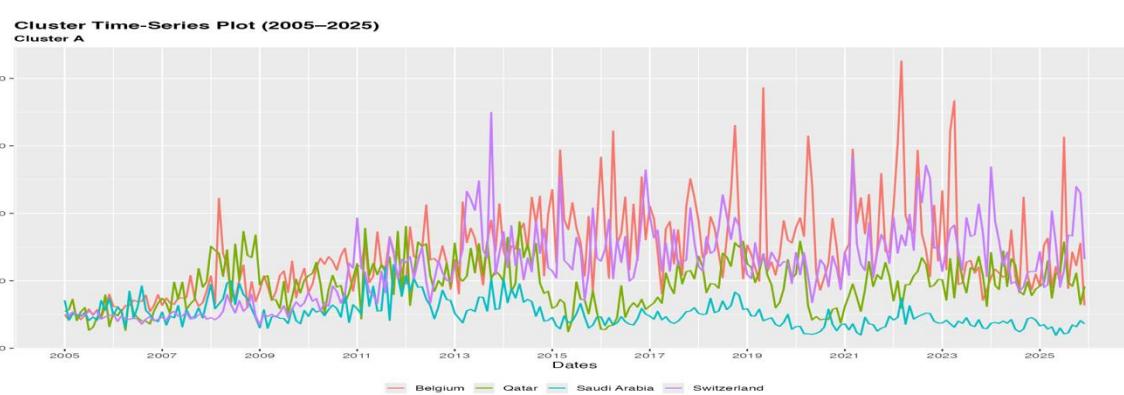
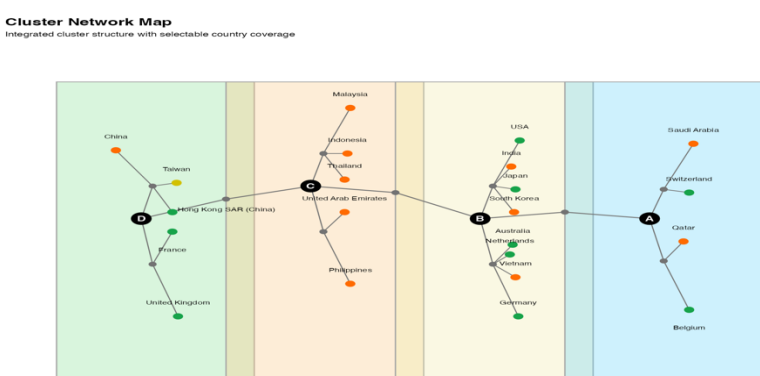
ACF (Autocorrelation)



The ACF plot indicates strong autocorrelation at lower lags, suggesting temporal dependency and justifying the use of time series modelling.

Cluster Analysis

Exploratory analysis highlights temporal trends, country-level differences, and seasonal patterns in trade data. Confirmatory analysis further validates these patterns through decomposition and autocorrelation. Overall, trade behaviour exhibits strong temporal structure and variability, supporting further modelling and clustering.



The **cluster network map** shows four clearly separated country groups, indicating that Singapore's trade partners can be organised into distinct structural clusters. It also shows that each cluster contains a different **mixture of developed, emerging, and newly developed economies**, suggesting that the grouping reflects similarity in trade behaviour.

The **time-series plot (use Cluster A as example)**, further shows that countries within the same cluster share broadly **similar long-term movement patterns**, although volatility and peak timing still differ across countries.

- Users can dynamically filter clusters to explore each cluster's patterns and trend varying with time

Objective

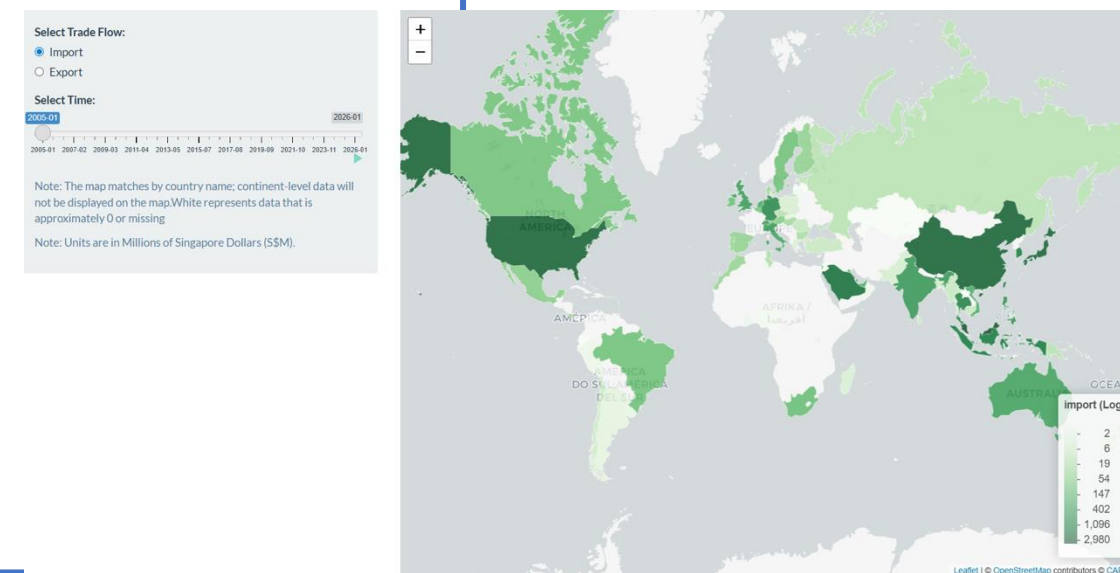
The Objective of this project is to :

- To examine Singapore's trade relationships across partner countries and uncover differences in trade scale, structure, and temporal dynamics.
- To apply visual analytics techniques, including exploratory and confirmatory time series analysis, clustering, and forecasting, to identify patterns and similarities among countries.
- To derive meaningful insights that support a more comprehensive understanding of Singapore's evolving global trade network.

Global Trade Dynamic Map

Description: This interactive choropleth map visualizes the geographical distribution of Singapore's trade volume globally. (Import/Export)

Analytical Purpose: To identify key trading hubs and observe the spatial-temporal evolution of import and export concentrations.



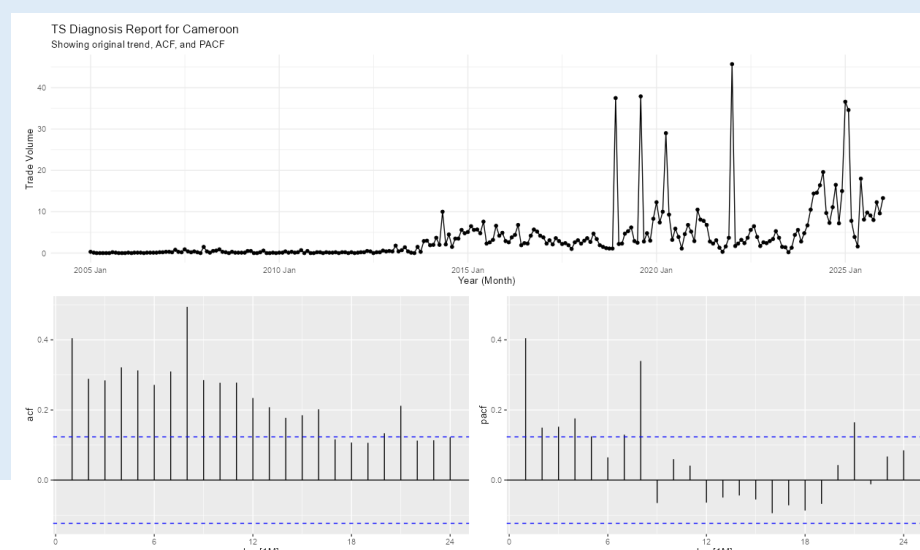
- It uses logarithmic scale to visualize colors, making low-level countries more prominent.
- The temporal slider allows for dynamic observation of shifts in international trade partnerships over time.

Time Series Diagnosis

Description: A comprehensive diagnostic panel performing time series decomposition and autocorrelation analysis. (Each country)

Trend Analysis: The top panel tracks historical trade fluctuations, highlighting periodicity and volatility.

ACF & PACF Plots: These are used to identify the underlying stochastic process and verify if the series is stationary or contains seasonal components.



Automated ETS Forecasting

Description:

- Automated selection result report for ETS model (Error, Trend, Seasonal), including: Error (M), Trend (N), Seasonal (M), Smoothing Parameters, and goodness of fit. (Each country)
- Future trade projections generated using the Automated Optimal ETS (Error, Trend, Seasonal) algorithm. (Each country).

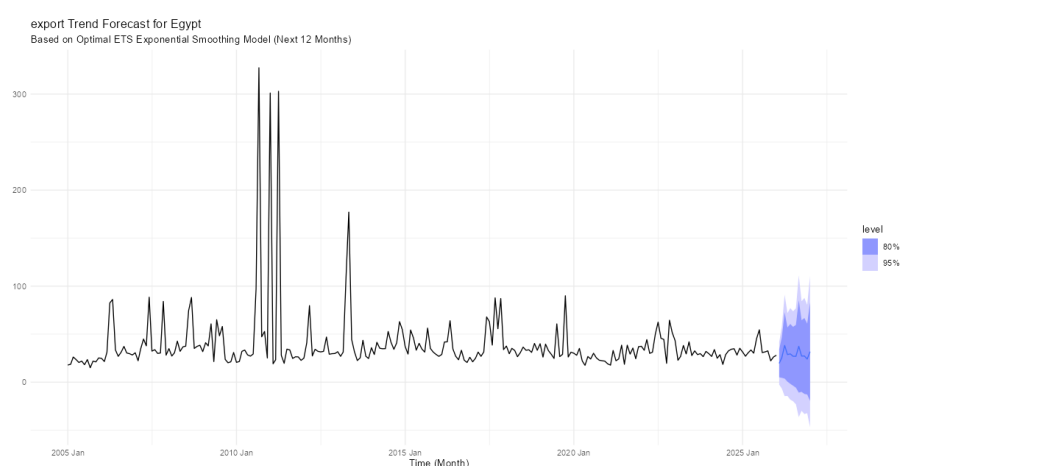
Analytical Purpose: To provide data-driven foresight for policy-makers and businesses, enabling more accurate strategic planning for the next 12 months.

```
Series: total
Model: ETS(M,N,M)
Smoothing parameters:
alpha = 0.4141512
beta = 0.00328913
gamma = 0.0001002293

Initial states:
l[0] s[0] s[1] s[2] s[3] s[4] s[5] s[6]
17.95092 1.184833 0.8368196 0.9490126 0.9486276 1.303136 0.9417226 0.9443988 1.032435
s[7] s[8] s[9] s[10] s[11]
1.009687 1.344048 0.8955269 0.6902287 1.104357

sigma^2: 0.3342

AIC AICc BIC
2971.795 2974.399 3031.863
```



Shiny App Application

We created four modules:

- EDA & CDA,
- Using DTW cluster package to conduct cluster Analysis of top 30 countries, cluster network map is to show detailed of each cluster, and time-series plot is to show the clusters trend through 2005-2025,
- Geospatial visualization: World trade heatmap,
- Time series analysis & Forecasting.

Conclusion & Future Work

By integrating **exploratory & confirmatory analysis, visualisation, forecasting and clustering**, the project offers a more comprehensive view of Singapore's trade partner landscape. The findings suggest that trade partners differ not only in scale, but also in their **structural roles and temporal behaviour**.

Overall, the project shows that **visual analytics can make complex trade data more interpretable and support deeper understanding** of Singapore's global trade relationships.

Future Work

- Incorporate **additional trade dimensions** to move beyond aggregate country-level patterns and support more nuanced interpretation of Singapore's trade relationships.
- Introduce **external explanatory factors**, such as policy changes, global economic shocks, or geopolitical events, to better explain observed trade patterns and cluster differences.
- Compare and validate results **using alternative analytical methods**, so as to assess the robustness of the patterns identified across different modules.